

- 13 A** Total number of moles of gas is constant. Since $PV = nRT$, we take the initial number of moles in the smaller (A) and larger bulb (B) to be n and $8n$ respectively. Total number of moles = $9n$.

At new equilibrium, the pressure will be the same for both.

$$P_f V = n' R T_A = n' R (80 + 273.15)$$

$$P_f (8)V = (9n - n') R T_B = (9n - n') R (10 + 273.15)$$

Dividing:

$$\frac{V}{8V} = \frac{n'(353.15)}{(9n - n')(283.15)}$$

$$\frac{9n - n'}{n'} = \frac{8 \times 353.15}{283.15}$$

$$\frac{9n}{n'} = \frac{8 \times 353.15}{283.15} + 1$$

$$n' = 0.8194n$$

$$\Delta n = n - 0.8194n = 0.18n$$